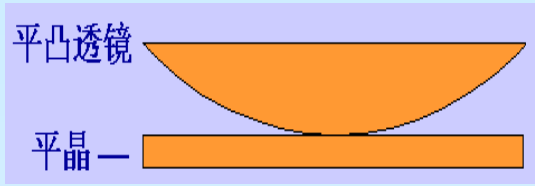
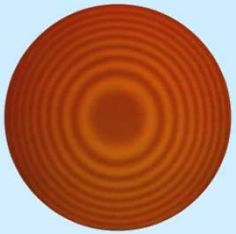
- 回顾：

2）牛顿环





$$2d_k+\frac{\lambda}{2}=\begin{cases}k\lambda & k=1,2\cdots\text{明}\\ (2k+1)\frac{\lambda}{2} & k=0,1,2\cdots\text{暗}\end{cases}$$

暗环半径：

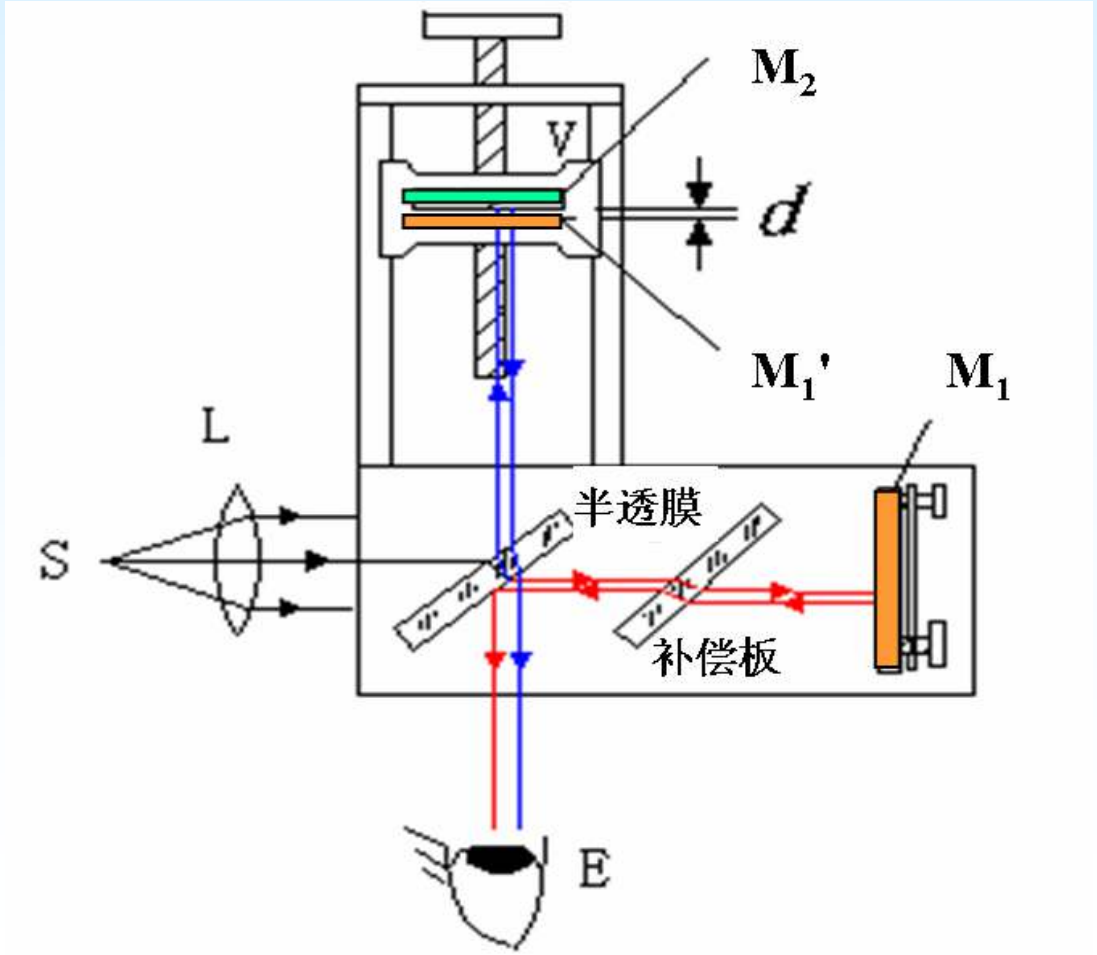
$$r_k=\sqrt{kR\lambda}\,(k=0,1\cdots)$$

迈克尔逊干涉仪

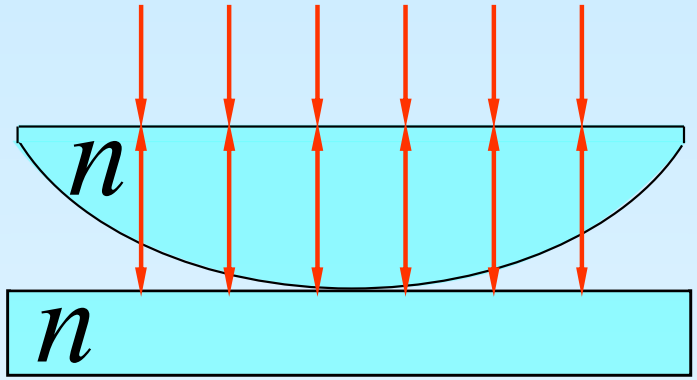
明纹 移动的数目 **N**

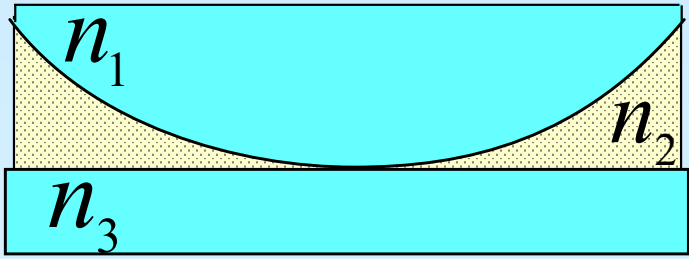
M_2平移的距离

$$\Delta d=N\frac{\lambda}{2}$$

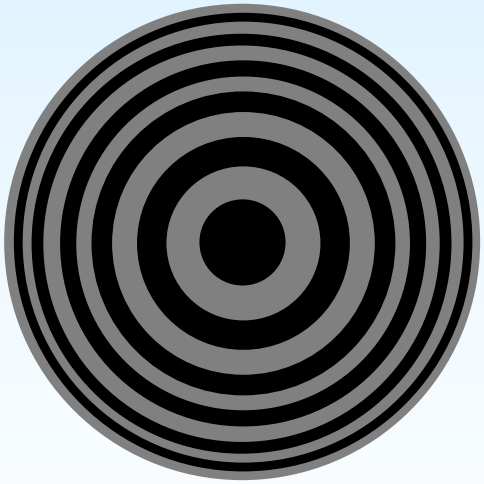


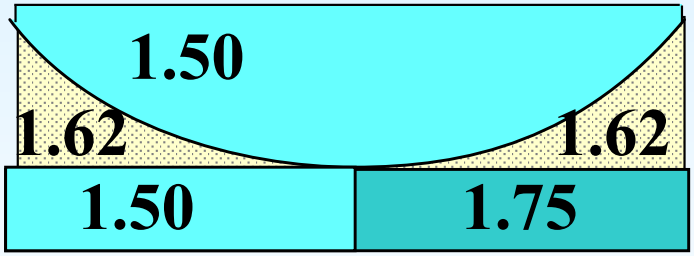
例1. 半波损失需具体问题具体分析。



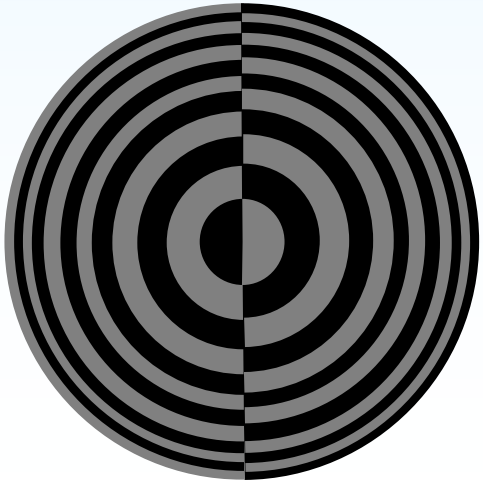


$$n_1 < n_2 < n_3$$









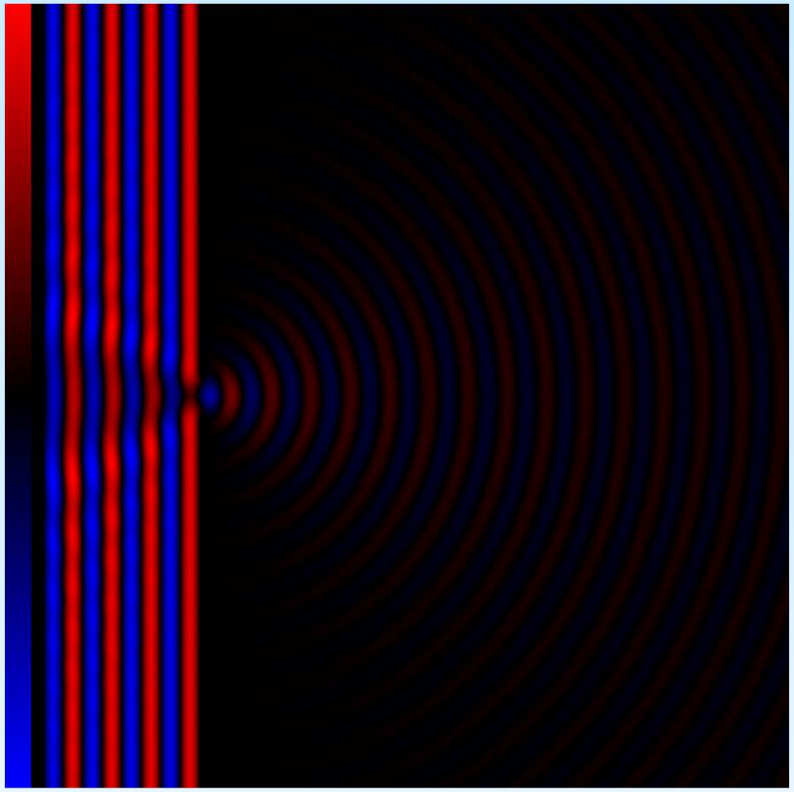
第11章

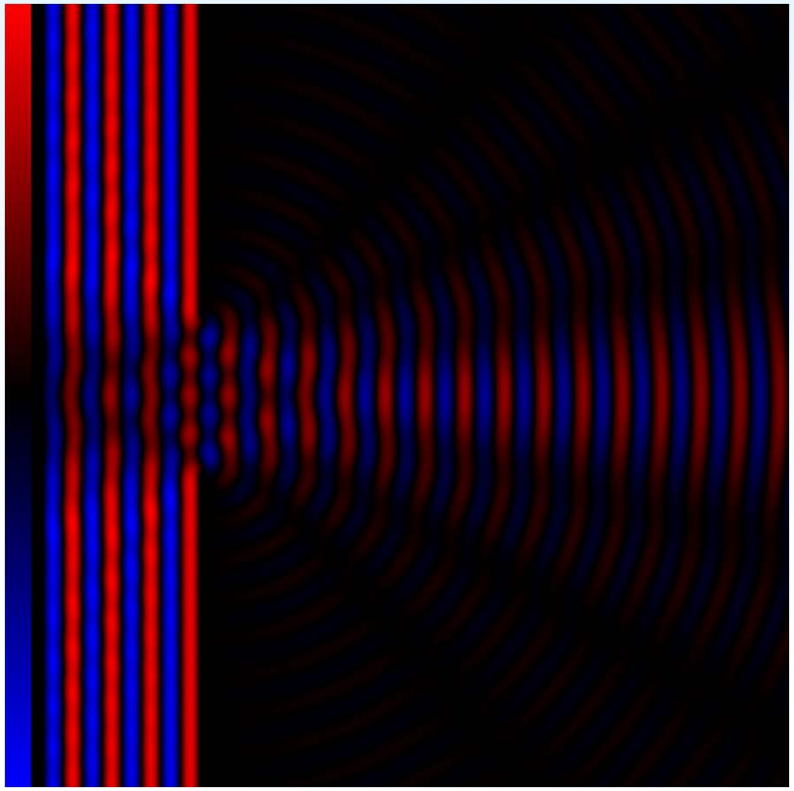
波动光学

第二部分

光的衍射

Diffraction Of Light



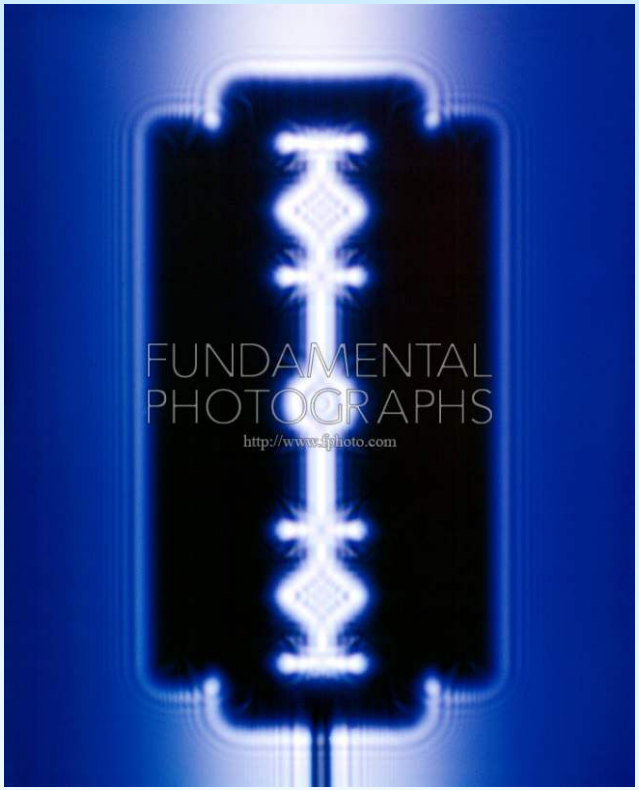


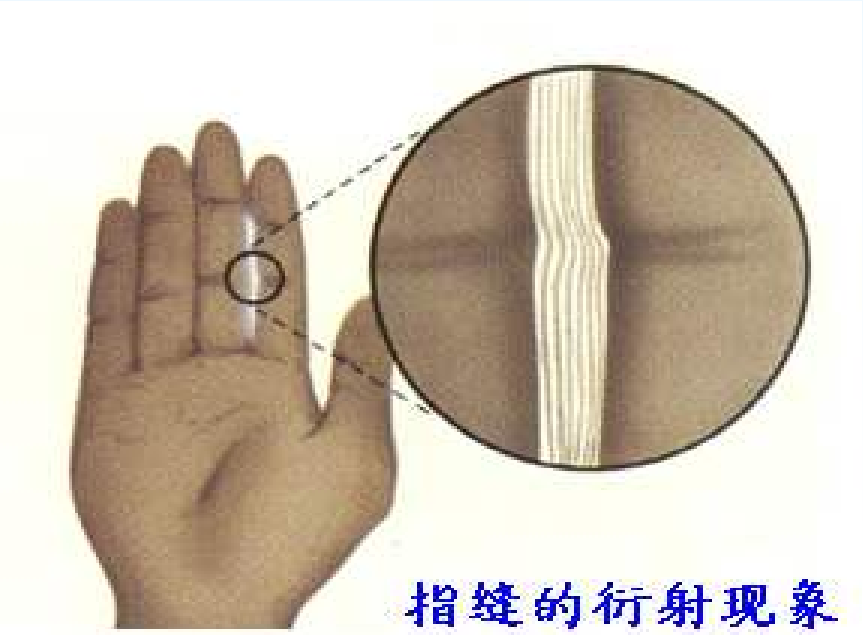


几种实物衍射图









指缝的衍射现象

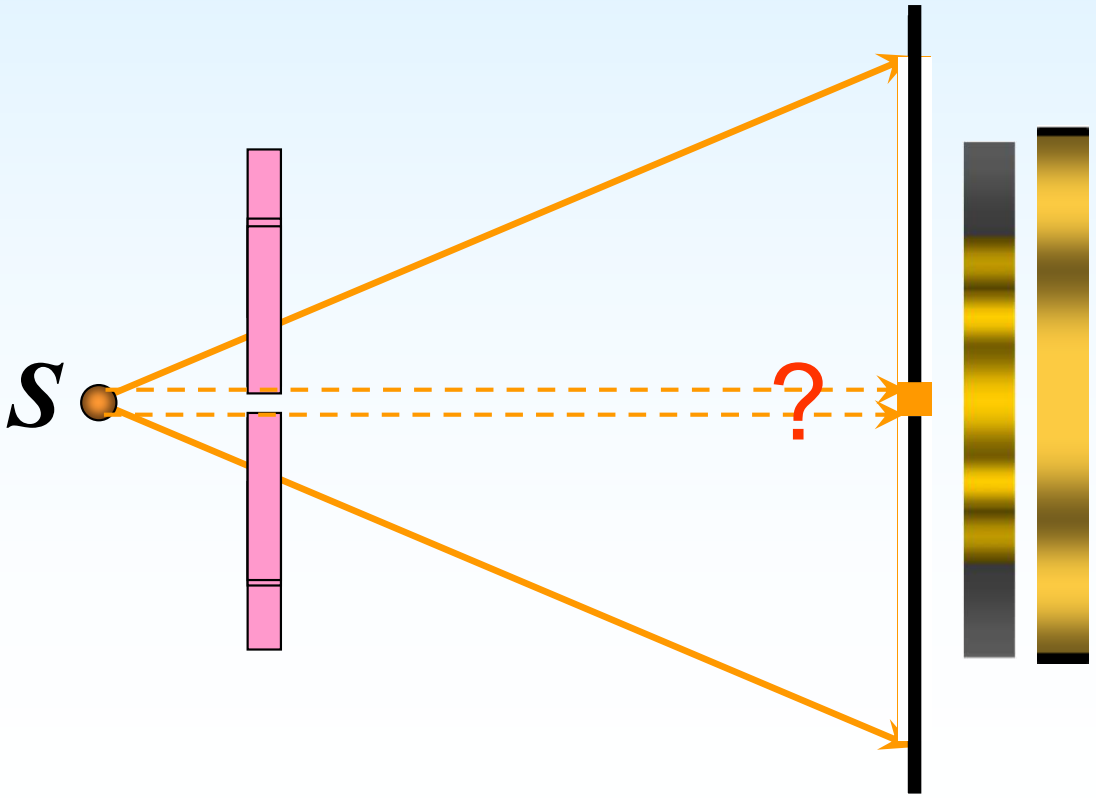
第5节 光波的衍射

一、光的衍射现象

波的衍射现象：波在传播过程中遇到障碍物，能够绕过障碍物的边缘而偏离直线传播的现象。

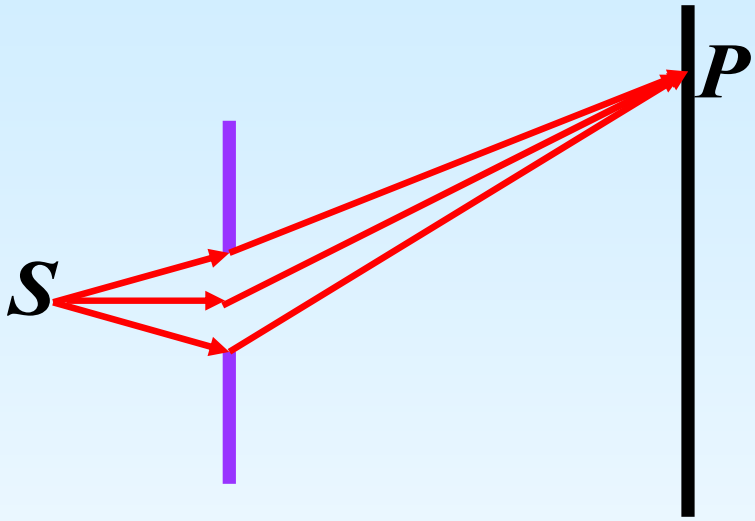
1.现象

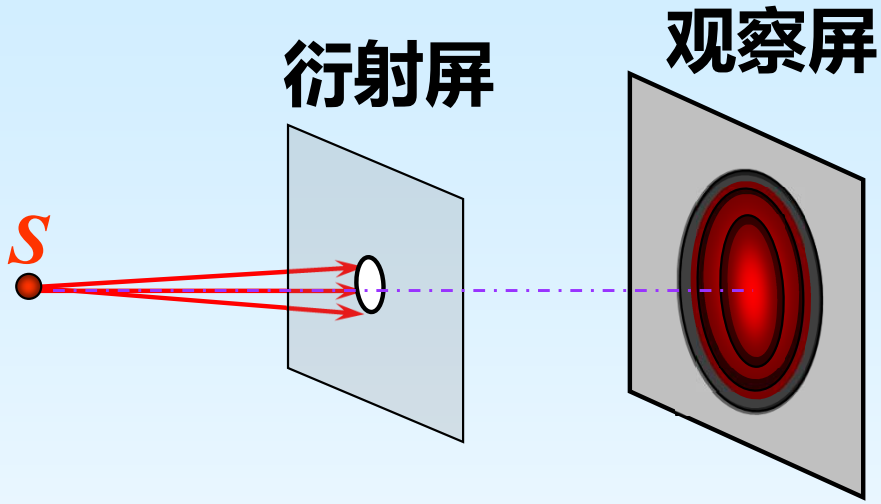
光在传播过程中遇到尺寸比光的波长大得不多的障碍物时，光会传到障碍物的阴影区并形成明暗变化的光强分布的现象。



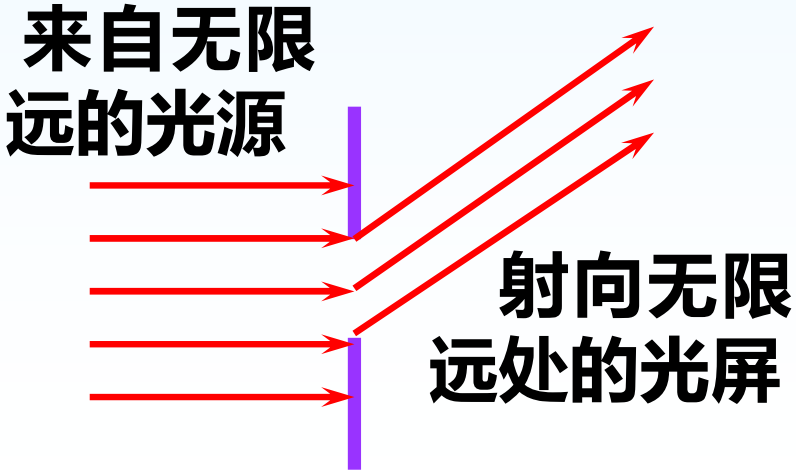
2.衍射分类

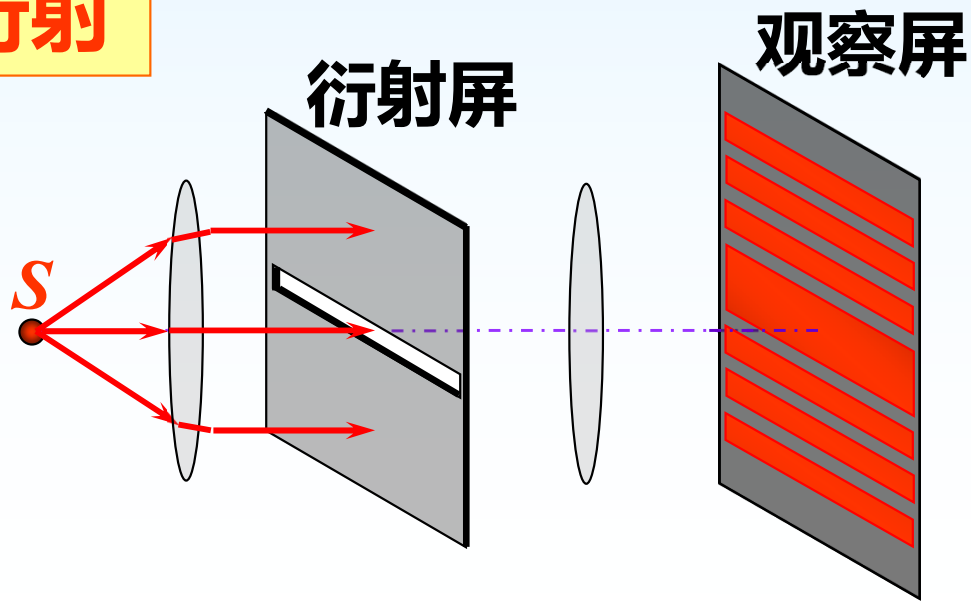
(1) 菲涅耳衍射　　近场衍射





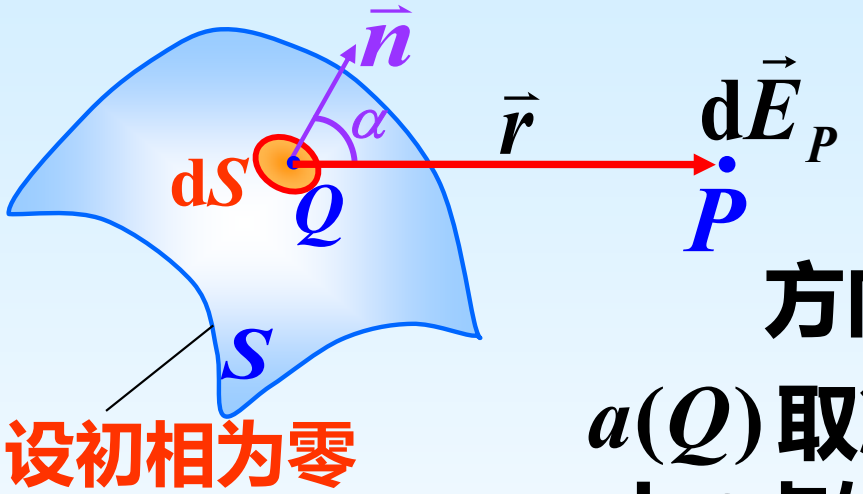
(2) 夫琅禾费衍射　　远场衍射





3.惠更斯——菲涅耳原理

波传到的任何一点都是子波的波源，各子波在空间某点的相干叠加，决定该点波的强度。



设初相为零

$$dE_P = dy \propto \frac{a(Q)f(\alpha)}{r}dS$$

方向因子$f(\alpha)$:
$$\begin{cases} \alpha = 0,\ f = f_{\max} \\ \alpha \uparrow \Rightarrow\ f(\alpha) \downarrow \\ \alpha \geq \dfrac{\pi}{2},\ f = 0 \end{cases}$$

$a(Q)$取决于波前上Q点处的强度

$$dy = c\frac{a(Q)f(\alpha)}{r}dS \cdot \cos(\omega t - \frac{2\pi r}{\lambda})$$ （c——比例系数）

$$y_P = c\int_S \frac{a(Q)f(\alpha)}{r}\cos(\omega t - \frac{2\pi r}{\lambda})dS = E_{0P}\cos(\omega t + \varphi_P)$$

菲涅耳衍射积分公式

P 处波的强度 $I_P \propto E_{0P}^2$

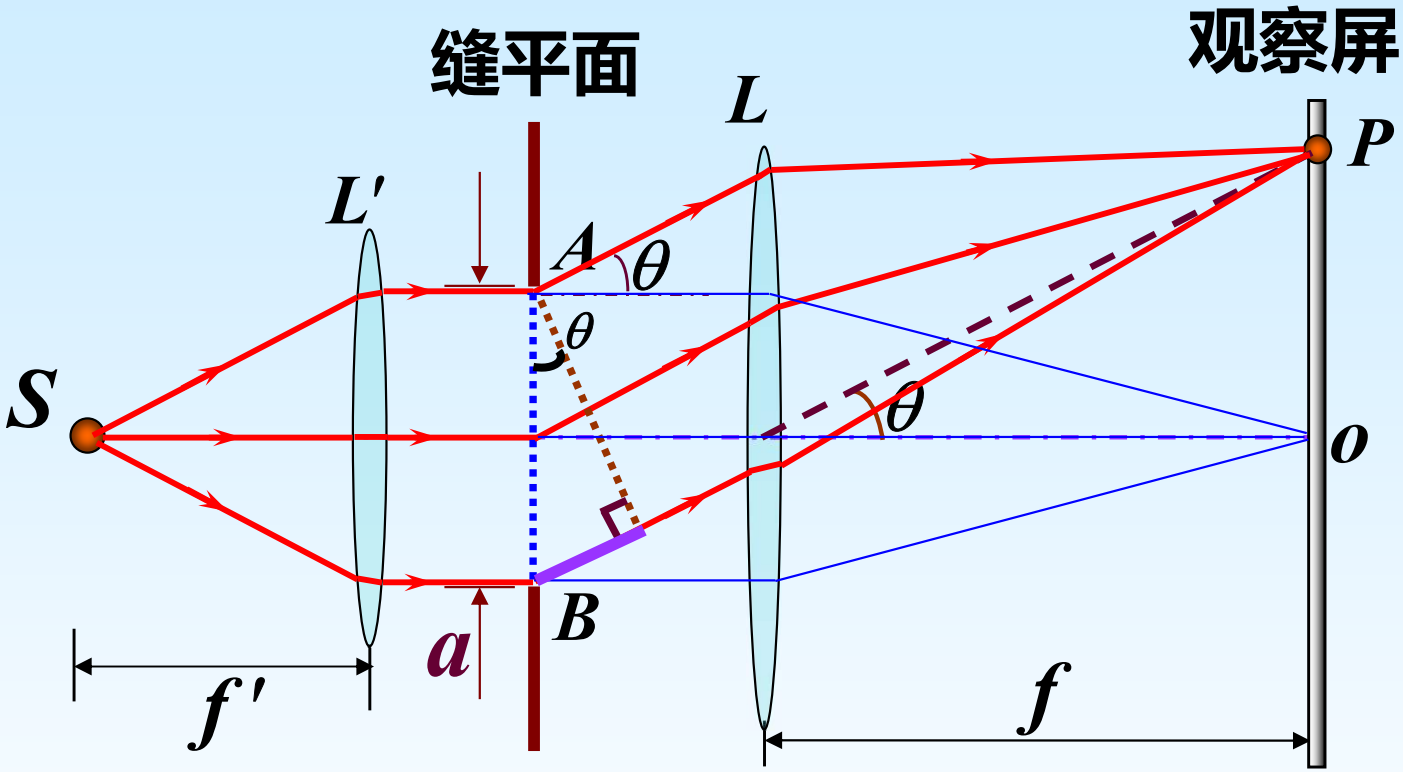
二、单缝夫琅和费衍射

1.装置

θ：衍射角

（向上为正，

向下为负）



2.衍射机制

屏上任一点的光振动为单缝所在处波面各子波源的相干光相干叠加的结果。

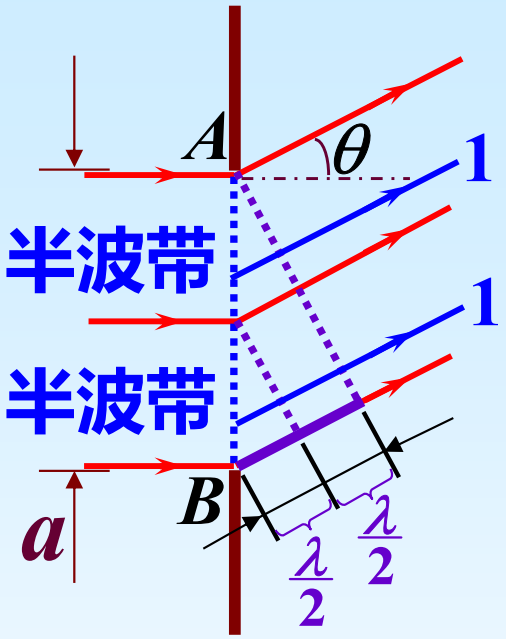
$\theta = 0$

中央明纹

单缝上下沿光线到P点的光程差为：$\delta = a\sin\theta$

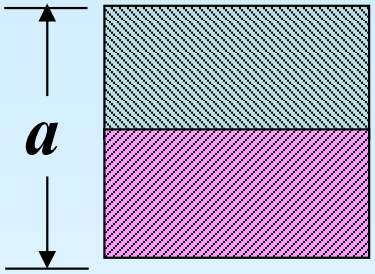
3.衍射光强公式

（一）半波带法



若 $k'=2$

$$a\sin\theta=\lambda$$



暗纹

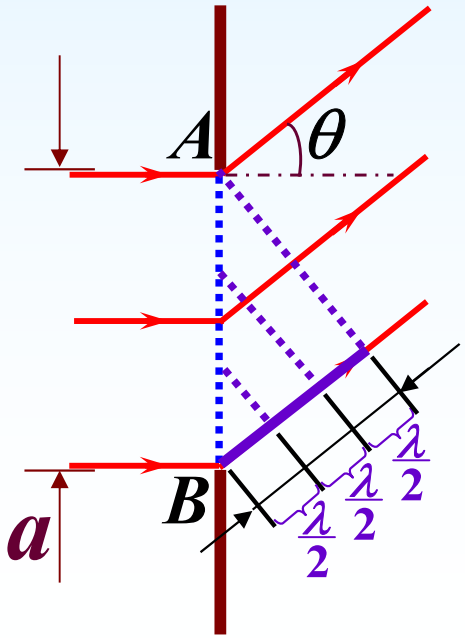
设：

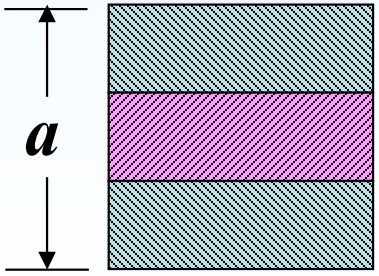
$$\delta=a\sin\theta=\pm k'\frac{\lambda}{2}$$

$$k'=1,2,3\cdots$$

若 $k'=3$

$$a\sin\theta=\frac{3}{2}\lambda$$





明纹

暗纹条件：

$$a\sin\theta=\pm k\lambda,\quad k=1,2,3\cdots$$

明纹条件：

$$a\sin\theta=\pm(2k+1)\frac{\lambda}{2},$$

$$k=1,2,3\cdots$$

中央明纹：$a\sin\theta=0$

(1)单缝衍射光强分布的特点：

1）条纹形状 θ角相同处光强相同　与狭缝平行的条纹

2）中央主极大

$\theta = 0$处，$I = I_0 = I_{\max}$　屏上正对狭缝中心的O点。

3）极小（暗纹）

$$a\sin\theta = \pm k\lambda \qquad k = 1,2,3\cdots$$

4）明条纹宽度　相邻暗纹间的角距离。$a\sin\theta = \pm k\lambda$

中央明纹： k=±1的两暗点之间范围

$$-\frac{\lambda}{a} < \sin\theta < \frac{\lambda}{a}$$

半角宽度 $\sin\theta_1 = \frac{\lambda}{a}$

第1级明纹： k=1、2的两暗点之间范围

$$\frac{\lambda}{a} < \sin\theta < \frac{2\lambda}{a}$$

通常：$\sin\theta_1 \approx \theta_1$ **中央明纹：**

角宽度

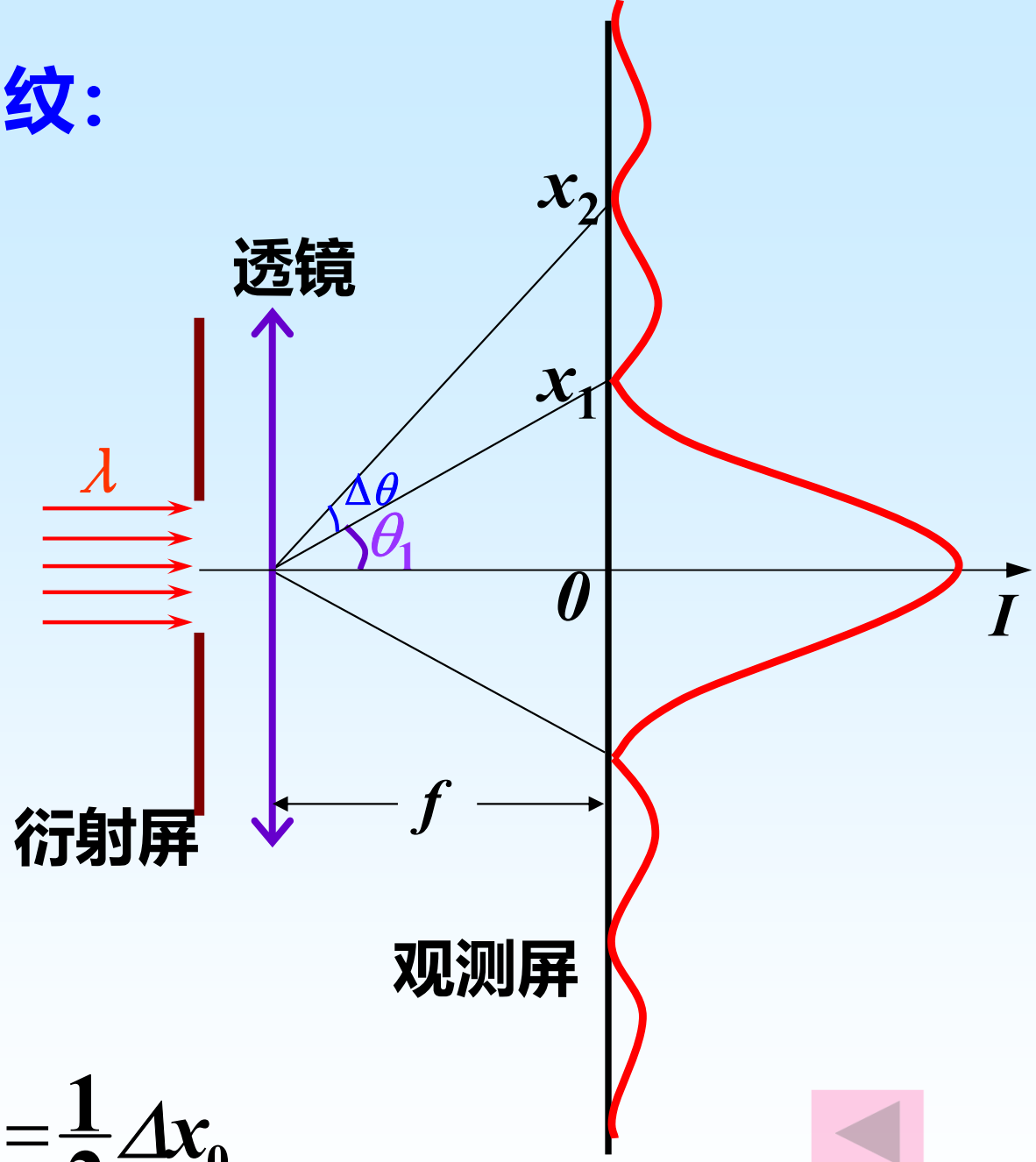
$$\Delta\theta_0 = 2\theta_1 \approx 2\frac{\lambda}{a}$$

线宽度

$$\Delta x_0 = 2f \cdot \mathrm{tg}\,\theta_1$$

$$\approx 2f\theta_1 = 2f\frac{\lambda}{a}$$

$$\Delta x_0 \propto \frac{\lambda}{a}$$

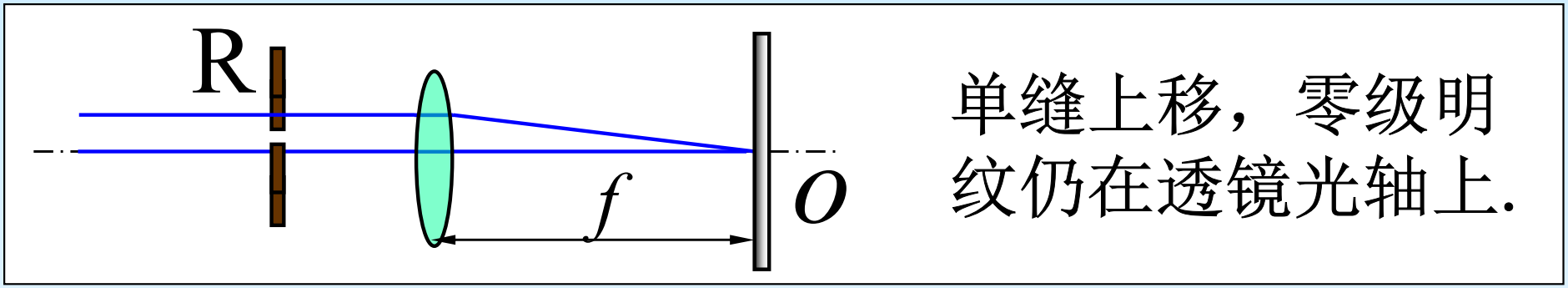


其它明纹： $\Delta x \approx \frac{f\lambda}{a} = \frac{1}{2}\Delta x_0$

次极大条纹的宽度是中央主极大宽度的一半。

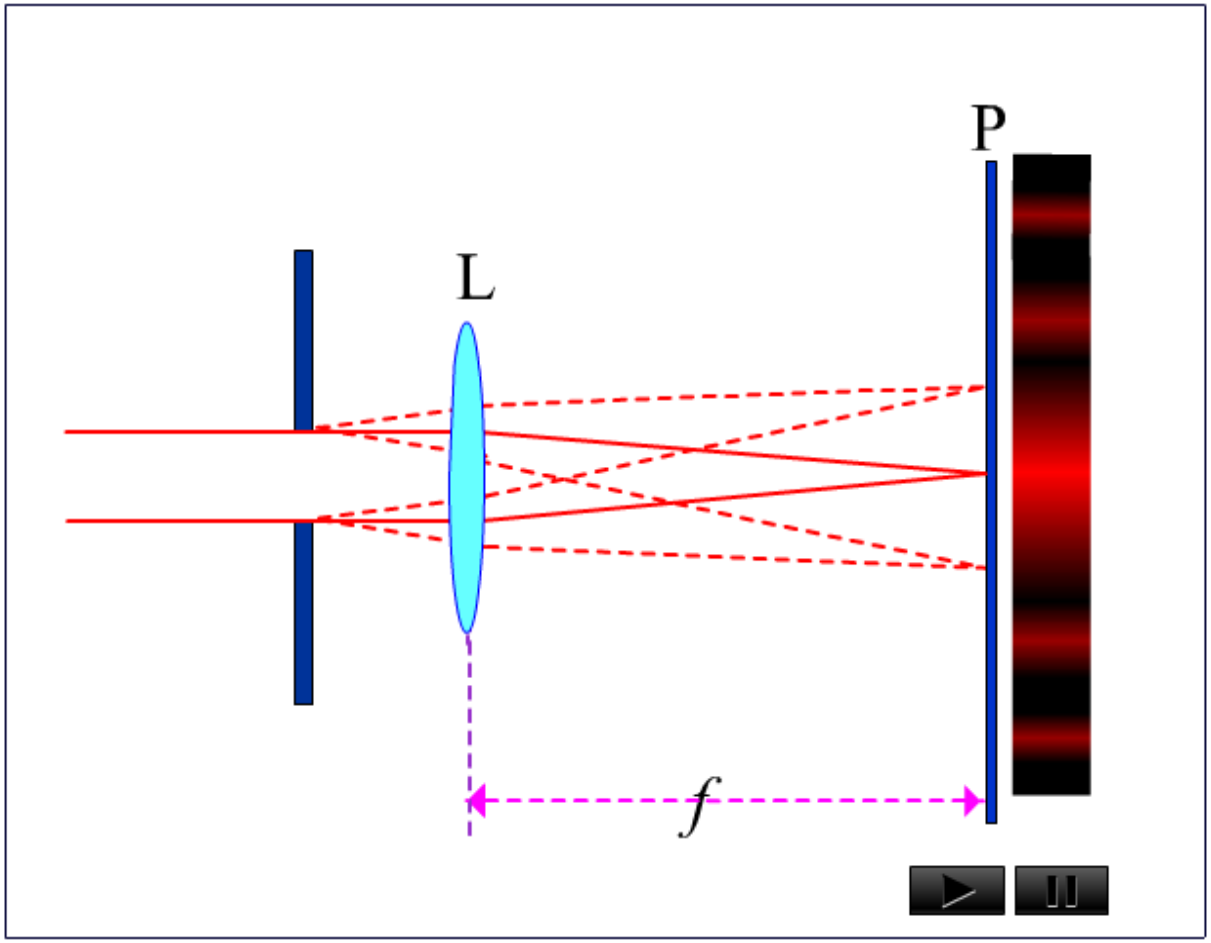
5）单缝衍射的动态变化

- 单缝上下移动，根据透镜成像原理衍射图不变 .



缝宽变化对条纹的影响：$\Delta x \propto \frac{1}{a}$

$$\Delta x_0 \propto \frac{\lambda}{a}$$



缝宽越小，条纹展得越开，衍射作用愈显著。

缝宽越大，条纹向中央明纹靠拢，衍射作用愈不显著。

当 $\frac{\lambda}{a} \to 0$ 时，$\Delta x \to 0$　只显出单一的明条纹

——单缝的几何光学像

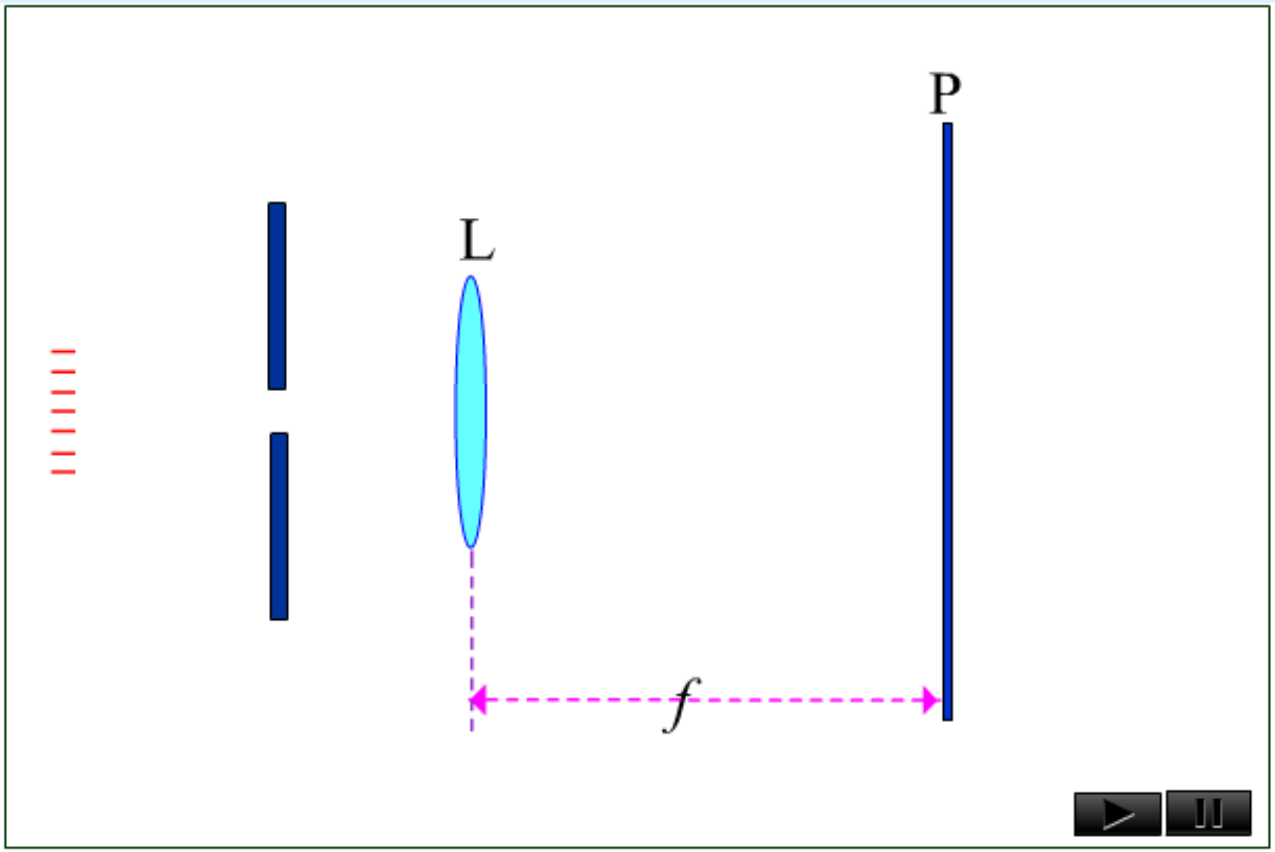
几何光学是波动光学在 $\lambda/a \to 0$ 时的极限情形

波长对条纹宽度的影响：$\Delta x \propto \lambda$

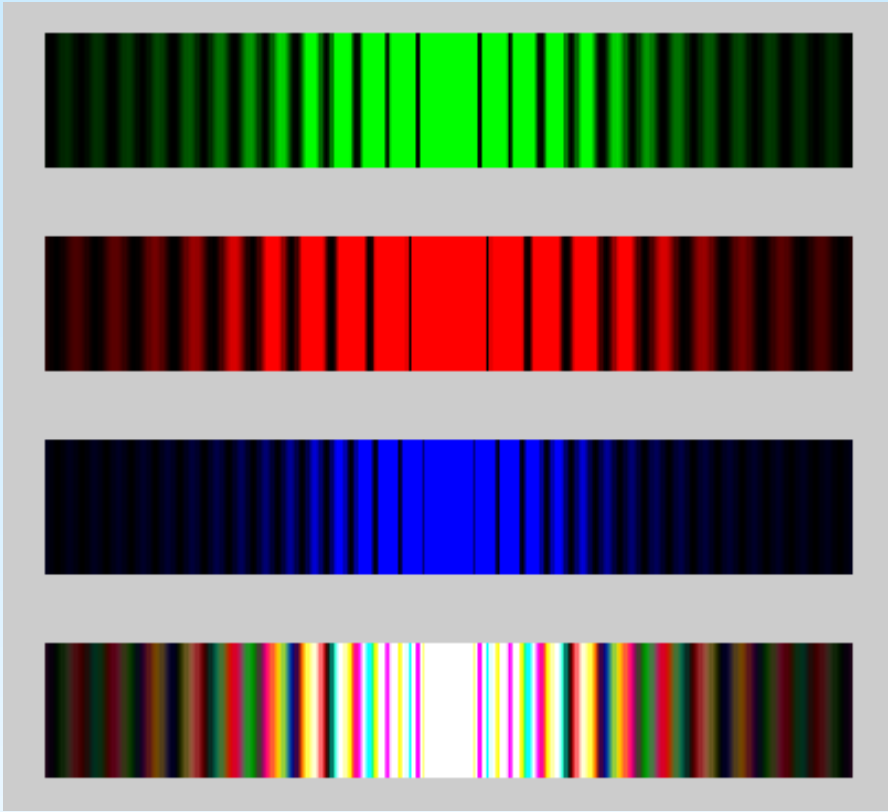
$$\Delta x_0 \propto \frac{\lambda}{a}$$

波长越长，条纹宽度越宽，衍射效应越明显。

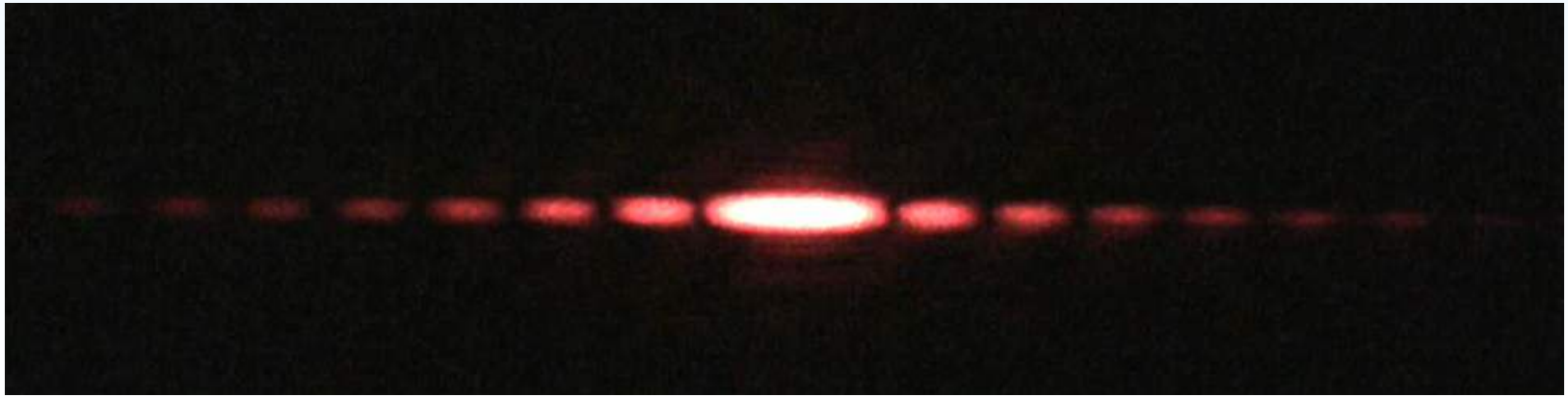
复色光（白光）入射：不同波长的光的明纹不完全重叠



屏幕中央O为白色明纹，最靠近O的为紫色，最远的为红色。



*S*为线光源时的衍射图样

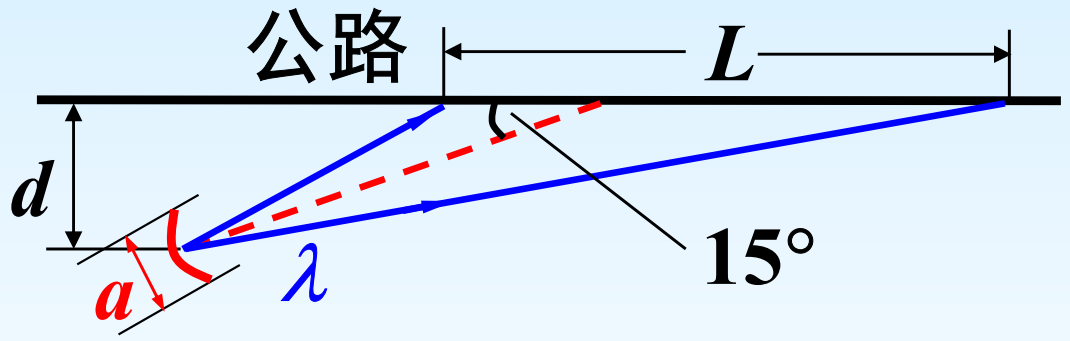


*S*为点光源时的衍射图样

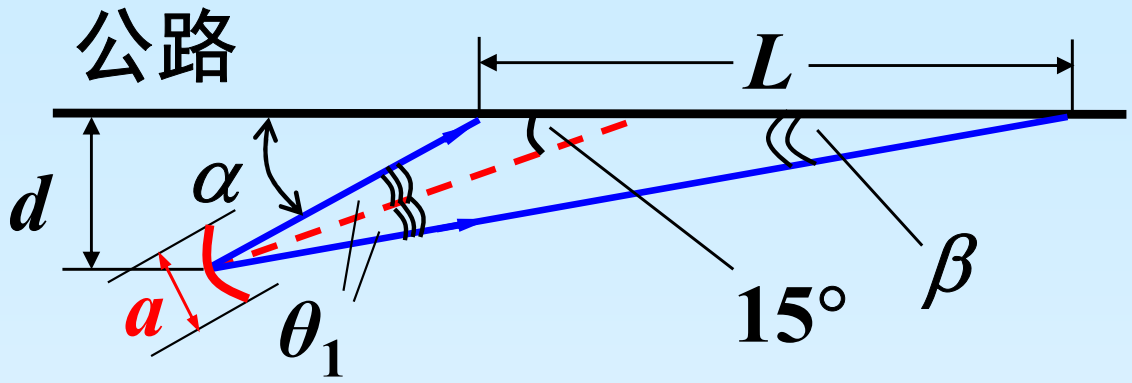
7）干涉和衍射的联系与区别

干涉：有限光束	}	相干	无本质	薄膜	纯干涉
衍射：无穷多子波	}	叠加	区别	双逢	干涉+衍射

例 如图所示：



已知：一波长为$\lambda = 30\text{mm}$的雷达在距离路边为$d = 15\text{m}$处，雷达射束与公路成15°角，天线宽度$a = 0.20\text{m}$。求雷达监视范围内公路的长度L。



解： 将雷达波束看成是单缝衍射的 0 级明纹

由角宽度 $\Delta\theta_0 = 2\theta_1 \approx 2\dfrac{\lambda}{a}$

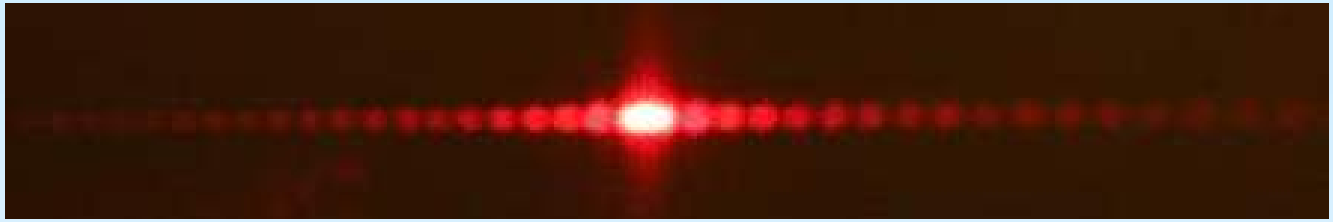
有 $$\sin\theta_1 = \frac{\lambda}{a} = \frac{30\,\text{mm}}{0.2\,\text{m}} = 0.15 \rightarrow \theta_1 \approx 8.63^\circ$$

如图 $\alpha = 15^\circ + \theta_1 = 23.63^\circ$ ， $\beta = 15^\circ - \theta_1 = 6.37^\circ$

所以 $$L = d(\text{ctg}\,\beta - \text{ctg}\,\alpha)$$

$$= 15(\text{ctg}\,6.37^0 - \text{ctg}\,23.63^0) \approx 100\,\text{m}$$

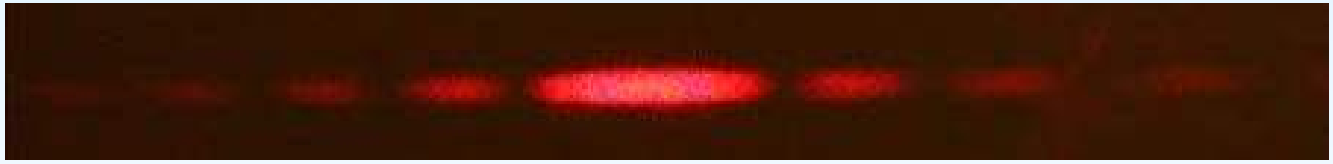
- 不同缝宽的单缝衍射条纹的实例比较



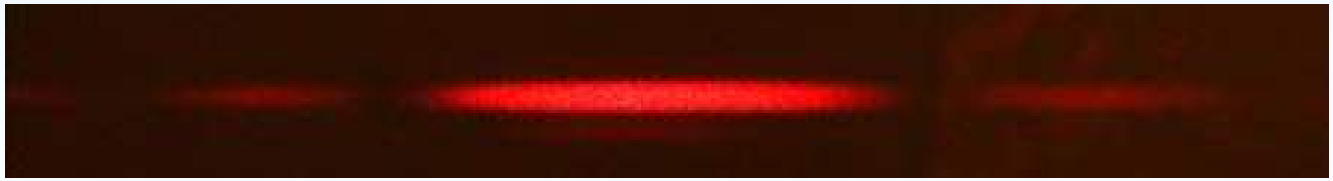
0.16 mm



0.08 mm

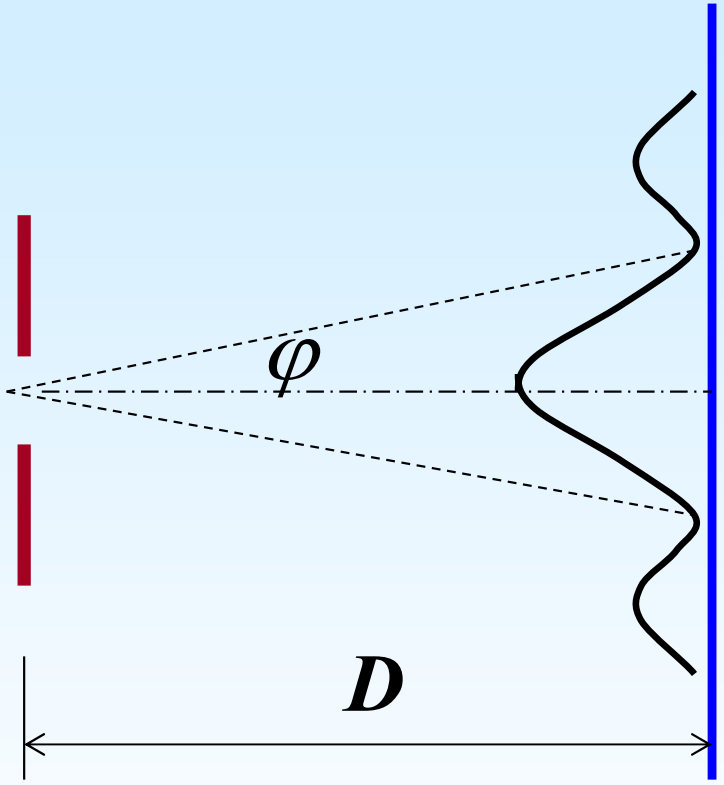


0.04 mm



0.02 mm

例题1 波长为 **546.0nm** 的平行光垂直照射在 $a = 0.437\text{mm}$ 的单缝上, 缝后有焦距为 **40cm** 的凸透镜, 求透镜焦平面上出现的衍射中央明纹的宽度.



解: $a\sin\varphi = \lambda$

$$\varphi \approx \sin\varphi = \frac{\lambda}{a}$$

$$L = 2x = 2D\cdot\tan\varphi$$

$$\approx 2D\varphi = \frac{2\lambda D}{a}$$

$$= \frac{2\times 5.460\times 10^{-7}\times 0.40}{0.437\times 10^{-3}} = 1.0\times 10^{-3}\,\text{m}$$

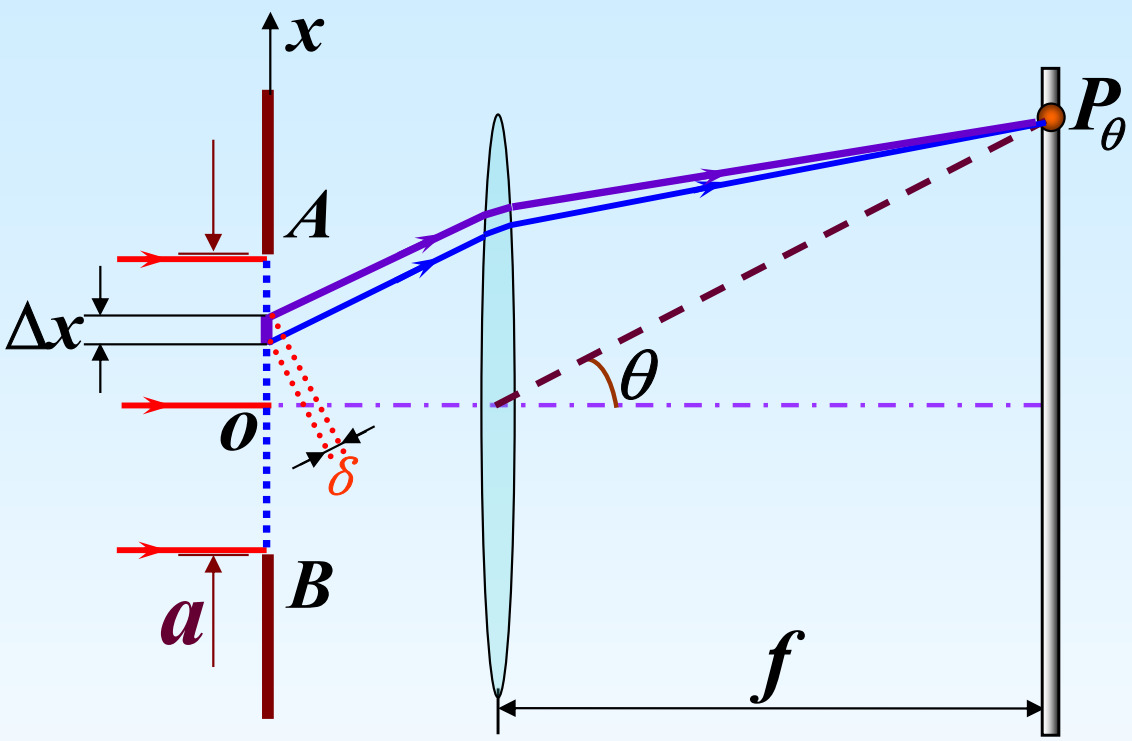
（二）振幅矢量法求光强公式（了解）

将单缝处波面分成N个窄条：

$$\Delta x = a/N$$

（N很大）

每个窄条发的子波在P点振幅近似相等，设为 A_0



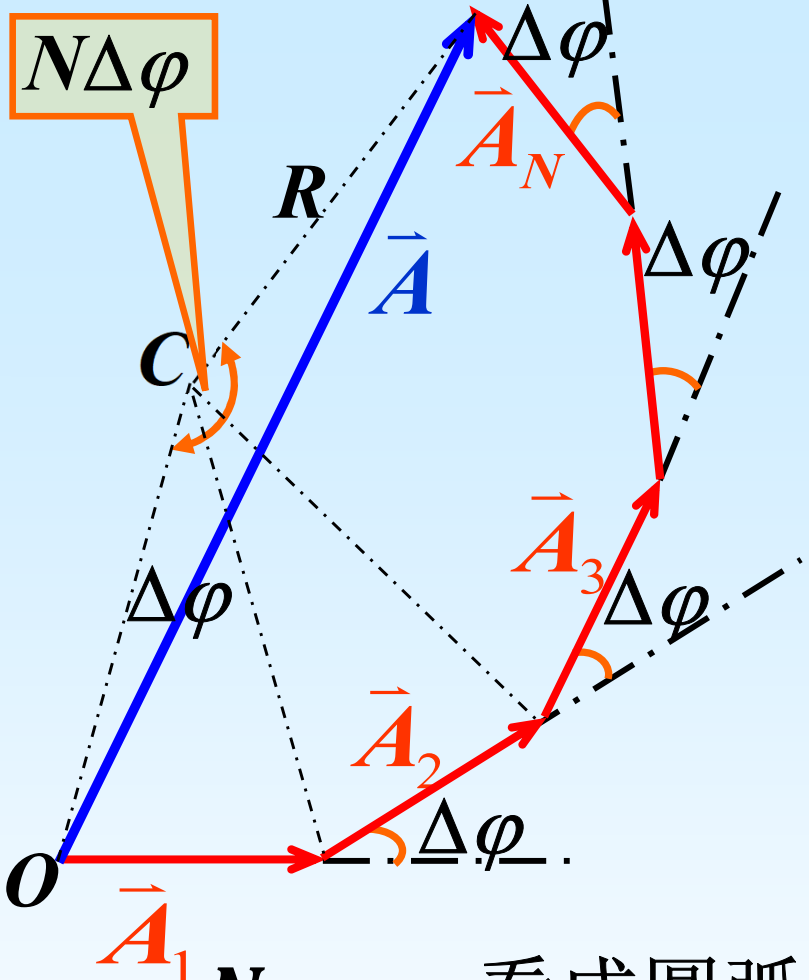
相邻窄条：$\Delta\varphi = \dfrac{2\pi}{\lambda}\cdot\Delta x\sin\theta = \dfrac{2\pi}{\lambda}\cdot\dfrac{a\sin\theta}{N}$

P处的合振幅E_P就是各窄条子波的振幅矢量和的模。

合成波振幅 $A = 2R\sin(\frac{N\Delta\varphi}{2})$

子波振幅 $A_1 = 2R\sin(\frac{\Delta\varphi}{2})$

$$A = A_1 \frac{\sin(\frac{N\Delta\varphi}{2})}{\sin(\frac{\Delta\varphi}{2})}$$



$N \to \infty$ 看成圆弧

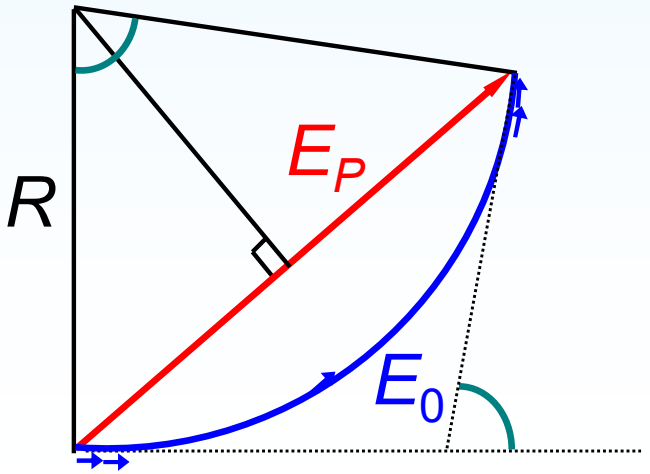
α

$$\Delta\varphi = \frac{2\pi}{\lambda}\cdot\frac{a\sin\theta}{N} = \frac{\pi a\sin\theta}{\lambda}\frac{2}{N}$$

$$A = A_0\frac{\sin(\alpha)}{\alpha}$$

$$I \propto A_{p'}^2 \quad I_0 \propto A_0^2$$

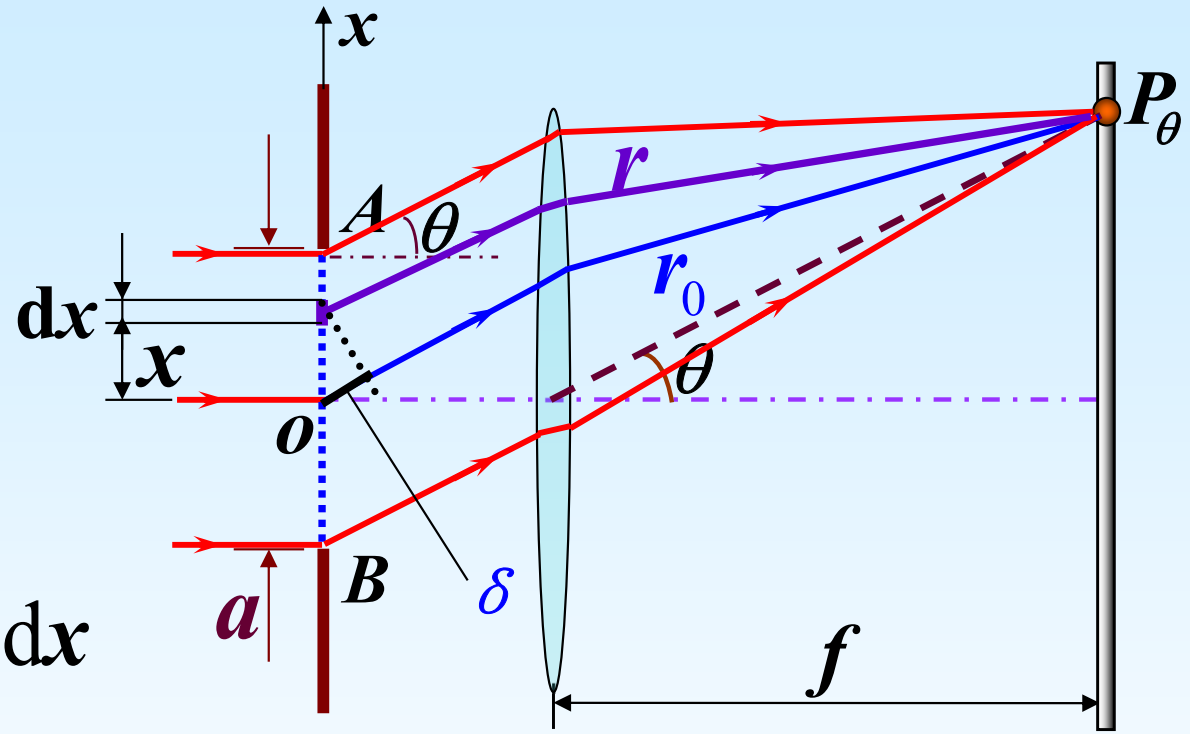
$$I = I_0(\frac{\sin\alpha}{\alpha})^2$$



(三)积分法 （了解）

$$\mathrm{d}y = c\frac{a(Q)f(\alpha)}{r}\mathrm{d}S\cos(\omega t - \frac{2\pi r}{\lambda})$$

$$c\frac{a(Q)f(\theta)}{r} \approx c'$$



dx宽的窄条波面在P处产生的振动为：

$$\mathrm{d}y_x = c''\cos 2\pi(\frac{t}{T} - \frac{r}{\lambda})\mathrm{d}x$$

x处与中心O处两子波源到P点的光程差为：$\delta = r - r_0 = -x\sin\theta$

单缝之受限波面各子波源引起P点的合振动为：

$$y = \int \mathrm{d}y_x = \int_{-\frac{a}{2}}^{\frac{a}{2}} c''\cos 2\pi(\frac{t}{T} - \frac{r_0 - x\sin\theta}{\lambda})\mathrm{d}x$$

$$y=\int_{-\frac{a}{2}}^{\frac{a}{2}} c''\cos 2\pi\left(\frac{t}{T}-\frac{r_0-x\sin\theta}{\lambda}\right)\mathrm{d}x$$

$$=c''\frac{\lambda}{2\pi\sin\theta}\int_{-\frac{a}{2}}^{\frac{a}{2}}\mathrm{d}\left[\sin 2\pi\left(\frac{t}{T}-\frac{r_0-x\sin\theta}{\lambda}\right)\right]$$

$$=c''\frac{\lambda}{2\pi\sin\theta}\cdot 2\cos 2\pi\left(\frac{t}{T}-\frac{r_0}{\lambda}\right)\sin\left(\frac{\pi a\sin\theta}{\lambda}\right)$$

$$=c''a\frac{\sin\left(\frac{\pi a\sin\theta}{\lambda}\right)}{\frac{\pi a\sin\theta}{\lambda}}\cdot\cos 2\pi\left(\frac{t}{T}-\frac{r_0}{\lambda}\right)$$

令：$\alpha=\dfrac{\pi a\sin\theta}{\lambda}$

电矢量

振幅： $E_\theta=E_0\dfrac{\sin\left(\frac{\pi a\sin\theta}{\lambda}\right)}{\frac{\pi a\sin\theta}{\lambda}}=E_0\dfrac{\sin\alpha}{\alpha}$

(1)单缝衍射光强公式：

$$\alpha = \frac{\pi a \sin\theta}{\lambda}$$

P_θ点的光强 $I = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2$

$\propto a^2$

单缝衍射因子

(2)单缝衍射光强分布的特点：

1）条纹形状

θ角相同处光强相同　与狭缝平行的相互平行条纹

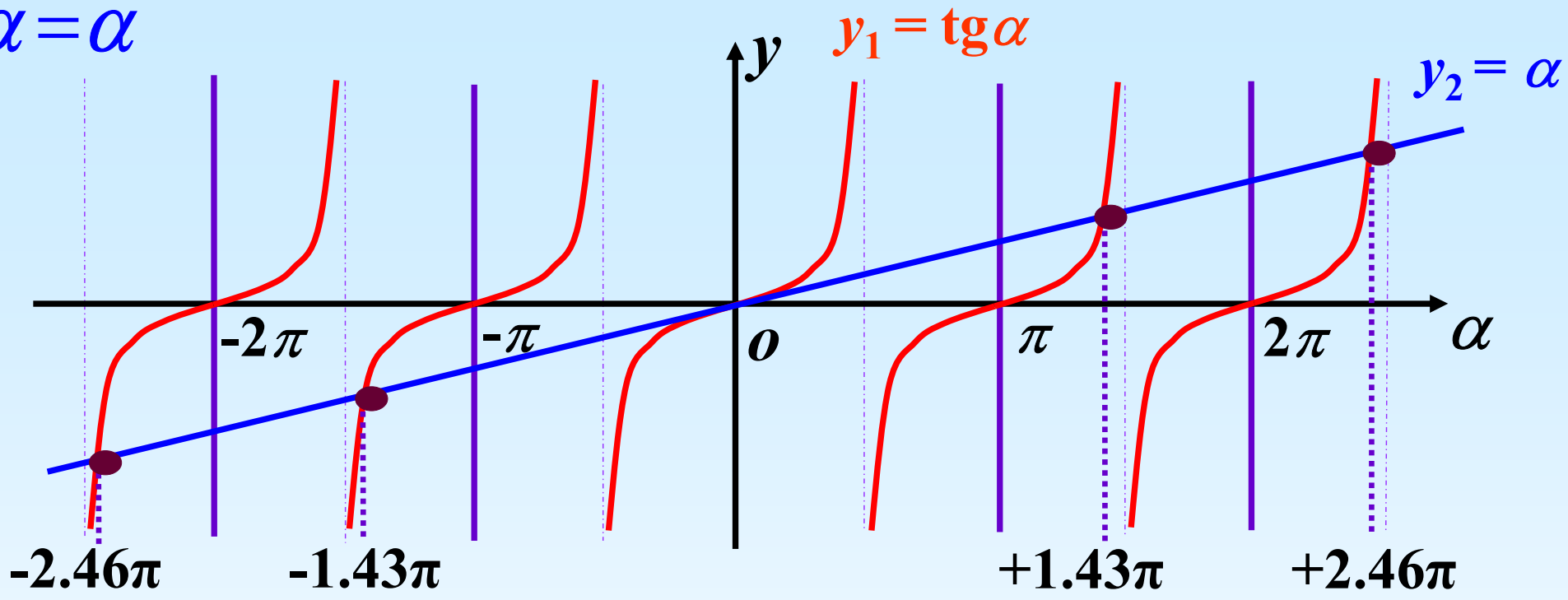
2）中央主极大

$\theta = 0$处，$I = I_0 = I_{\max}$　屏上正对狭缝中心的O点。

中央明纹、主极大、零级衍射斑

3）次极大　$\frac{\mathrm{d}}{\mathrm{d}\alpha}\left(\frac{\sin\alpha}{\alpha}\right)^2 = 0$　$\mathrm{tg}\,\alpha = \alpha$　图解：

$\mathrm{tg}\alpha = \alpha$



$$\alpha = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \cdots$$

$$a\sin\theta = \pm 1.43\lambda, \pm 2.46\lambda, \pm 3.47\lambda, \cdots$$

$$a\sin\theta = \pm(2k+1)\frac{\lambda}{2}$$

4）极小（暗纹）

——衍射次极大

$\alpha = \pm k\pi$，　$k = 1,2,3\cdots$时，$\sin\alpha = 0 \rightarrow I = 0$

$$a\sin\theta = \pm k\lambda \qquad k = 1,2,3\cdots$$

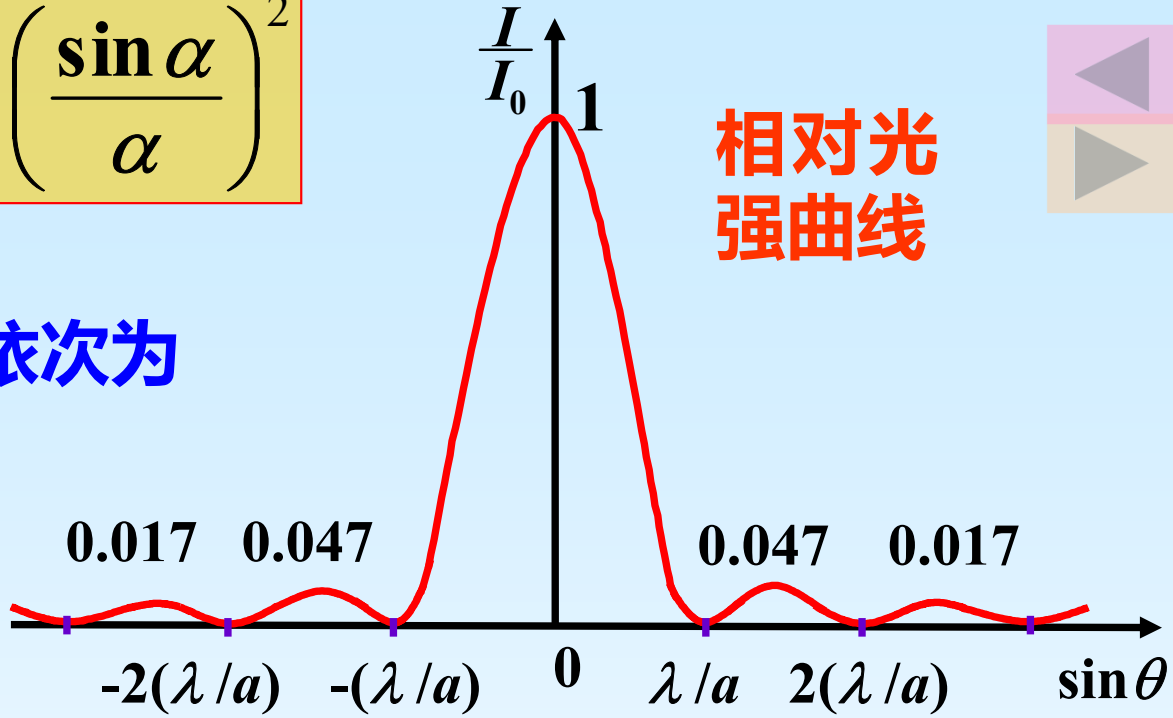
5）光强分布曲线 $I = I_0\left(\dfrac{\sin\alpha}{\alpha}\right)^2$

中央主极大: $I_{max} = I_0$

次极大：中央往外各次极大依次为

$0.0472I_0$，$0.0165I_0$，$0.0083I_0$，…

$I_{次极大} << I_{主极大}$



即单缝衍射光强集中在中央零级明条纹处。

6）明条纹宽度 **相邻暗纹间的角距离。** $a\sin\theta = \pm k\lambda$

中央明纹： k=±1的两暗点之间范围 $-\dfrac{\lambda}{a} < \sin\theta < \dfrac{\lambda}{a}$ **半角宽度** $\sin\theta_1 = \dfrac{\lambda}{a}$

第1级明纹： k=1、2的两暗点之间范围 $\dfrac{\lambda}{a} < \sin\theta < \dfrac{2\lambda}{a}$